Q.Create a package CIE which has two classes- Student and Internals. The class Personal has members like usn, name, sem. The class internals has an array that stores the internal marks scored in five courses of the current semester of the student. Create another package SEE which has the class External which is a derived class of Student. This class has an array that stores the SEE marks scored in five courses of the current semester of the student. Import the two packages in a file that declares the final marks of n students in all five courses.

Sudent.java

```
package CIE;

public class Student {
   public String usn;
   public String name;
   public int sem;

   public Student() {}

   public Student(String usn, String name, int sem) {
      this.usn = usn;
      this.name = name;
      this.sem = sem;
   }

   public void displayDetails() {
      System.out.println("USN: " + usn);
      System.out.println("Name: " + name);
      System.out.println("Semester: " + sem);
   }
}
```

External.java

```
package CIE;
import java.util.Scanner;

public class Internals extends Student {
   public int imarks[] = new int[5];
   Scanner sc = new Scanner(System.in);

   public void setImarks() {
      System.out.println("Enter 5 internal marks:");
      for (int i = 0; i < 5; i++)
```

```
            imarks[i] = sc.nextInt();
    }

    public void displayImarks() {
        for (int i = 0; i < 5; i++)
            System.out.println("Subject " + (i + 1) + ": " + imarks[i]);
    }
}
```

External.java

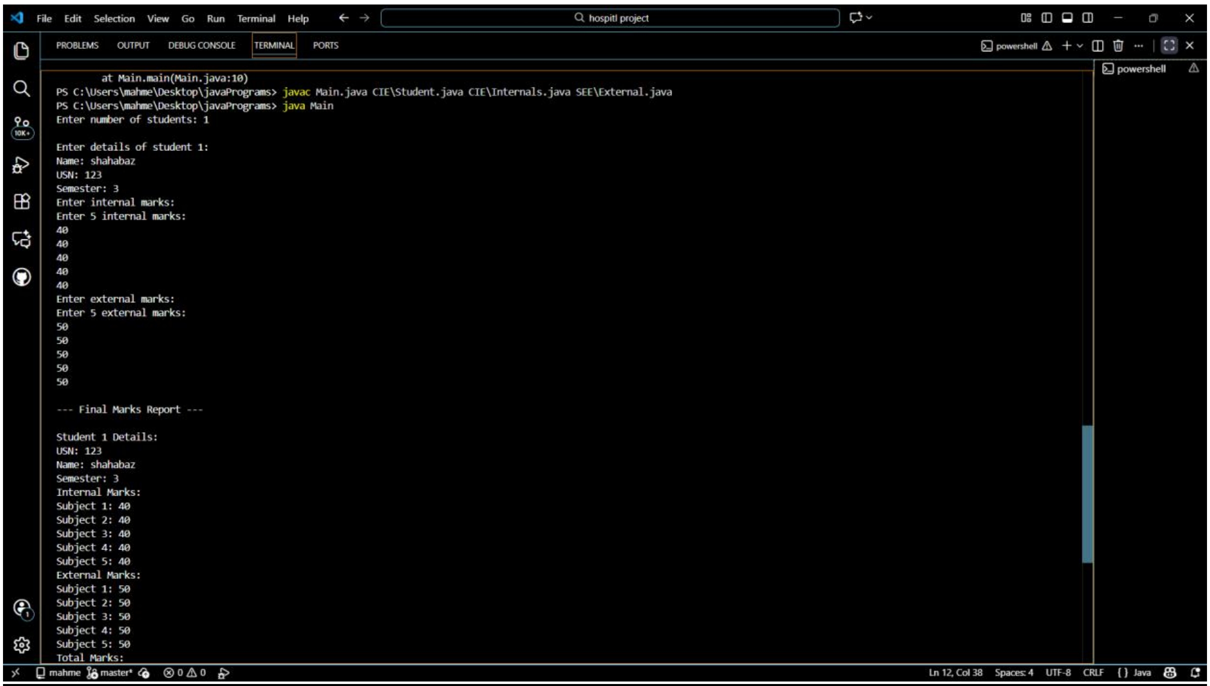
```
package SEE;
import java.util.Scanner;
import CIE.Student;

public class External extends Student {
    public int emarks[] = new int[5];
    Scanner sc = new Scanner(System.in);


    public void setEmarks() {
        System.out.println("Enter 5 external marks:");
        for (int i = 0; i < 5; i++)
            emarks[i] = sc.nextInt();
    }


    public void displayEmarks() {
        for (int i = 0; i < 5; i++)
            System.out.println("Subject " + (i + 1) + ": " + emarks[i]);
    }
}
```

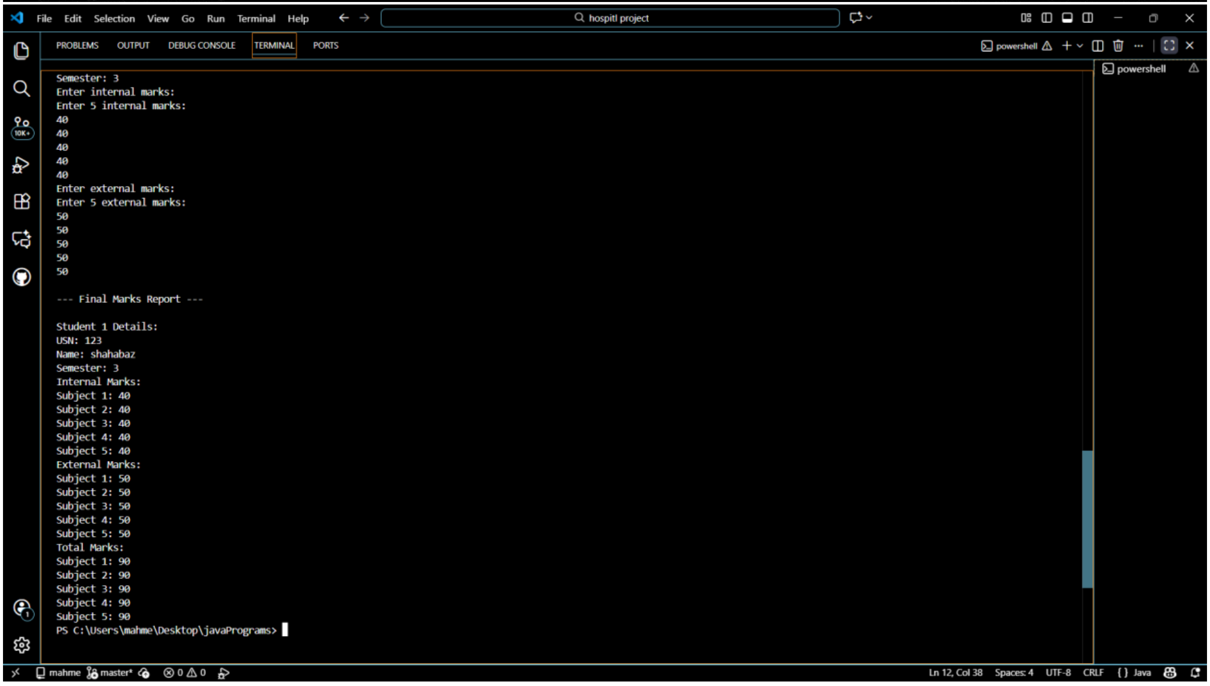
Main.java

```
import CIE.Student;
import CIE.Internals;
import SEE.External;
import java.util.Scanner;

public class Main {
```

```
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of students: ");
        int n = sc.nextInt();

        Student[] s = new Student[n];
        Internals[] i = new Internals[n];
        External[] e = new External[n];

        for (int j = 0; j < n; j++) {
            sc.nextLine();
            System.out.println("\nEnter details of student " + (j + 1) + ":");
            System.out.print("Name: ");
            String name = sc.nextLine();
            System.out.print("USN: ");
            String usn = sc.nextLine();
            System.out.print("Semester: ");
            int sem = sc.nextInt();

            s[j] = new Student(usn, name, sem);
            i[j] = new Internals();
            e[j] = new External();

            System.out.println("Enter internal marks:");
            i[j].setImarks();

            System.out.println("Enter external marks:");
            e[j].setEmarks();
        }

        System.out.println("\n Final Marks Report");
        for (int j = 0; j < n; j++) {
            System.out.println("\nStudent " + (j + 1) + " Details:");
            s[j].displayDetails();

            System.out.println("Internal Marks:");
            i[j].displayImarks();
            System.out.println("External Marks:");
            e[j].displayEmarks();

            System.out.println("Total Marks:");
            for (int k = 0; k < 5; k++)
                System.out.println("Subject " + (k + 1) + ": " + (i[j].imarks[k] + e[j].emarks[k]));
        }
    }
}
```

OUTPUT:-


hospitl project
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
at Main.main(Main.java:10)
PS C:\Users\mahme\Desktop\javaPrograms> javac Main.java CIE\Student.java CIE\Internals.java SEE\External.java
PS C:\Users\mahme\Desktop\javaPrograms> java Main
Enter number of students: 1
Enter details of student 1:
Name: shahabaz
USN: 123
Semester: 3
Enter internal marks:
Enter 5 internal marks:
40
40
40
40
40
Enter external marks:
Enter 5 external marks:
50
50
50
50
50
--- Final Marks Report ---
Student 1 Details:
USN: 123
Name: shahabaz
Semester: 3
Internal Marks:
Subject 1: 40
Subject 2: 40
Subject 3: 40
Subject 4: 40
Subject 5: 40
External Marks:
Subject 1: 50
Subject 2: 50
Subject 3: 50
Subject 4: 50
Subject 5: 50
Total Marks:
Ln 12, Col 38
Spaces: 4
UTF-8
CRLF
Java


hospitl project
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
Semester: 3
Enter internal marks:
Enter 5 internal marks:
40
40
40
40
40
Enter external marks:
Enter 5 external marks:
50
50
50
50
50
--- Final Marks Report ---
Student 1 Details:
USN: 123
Name: shahabaz
Semester: 3
Internal Marks:
Subject 1: 40
Subject 2: 40
Subject 3: 40
Subject 4: 40
Subject 5: 40
External Marks:
Subject 1: 50
Subject 2: 50
Subject 3: 50
Subject 4: 50
Subject 5: 50
Total Marks:
Subject 1: 90
Subject 2: 90
Subject 3: 90
Subject 4: 90
Subject 5: 90
PS C:\Users\mahme\Desktop\javaPrograms>
Ln 12, Col 38
Spaces: 4
UTF-8
CRLF
Java